

A PromiseBQP-complete String Rewriting Problem

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Challenges in Quantum Computer Science

- designing new efficient quantum algorithms
- understanding the computational power and limitations of quantum computers in comparison to those of classical computers

Language/Decision Problem

A language L is a subset of $\{0, 1\}^*$.

The corresponding decision problem (“language recognition”) consists in determining if $x \in L$ or $x \notin L$ for given $x \in \{0, 1\}^*$.

Complexity Class BQP

BQP is the class of decision problems that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

Let $\mathbf{x}$ be a binary string of length n . Then the application of U_n to $|\mathbf{x}, 0\rangle$ produces the state

$$U_n|\mathbf{x}, 0\rangle = \alpha_{\mathbf{x},0}|0\rangle \otimes |\psi_{\mathbf{x},0}\rangle + \alpha_{\mathbf{x},1}|1\rangle \otimes |\psi_{\mathbf{x},1}\rangle$$

such that

• $|\alpha_{\mathbf{x},1}|^2 \geq 2/3$ if $x \in L$

• $|\alpha_{\mathbf{x},1}|^2 \leq 1/3$ if $x \notin L$

Promise Problem

A promise problem Π is a pair $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$ of non-intersecting sets, i.e., $\Pi_{\text{YES}}, \Pi_{\text{NO}} \subseteq \{0, 1\}^*$ and $\Pi_{\text{YES}} \cap \Pi_{\text{NO}} = \emptyset$.

The set $\Pi_{\text{YES}} \cup \Pi_{\text{NO}}$ is called the *promise*.

Standard “language recognition” problems can be cast as the special case in which that promise is the set of all strings $\Pi_{\text{YES}} \cup \Pi_{\text{NO}} = \{0, 1\}^*$ (*trivial promise*)

Complexity Class PromiseBQP

PromiseBQP is the class of promise problems $(\Pi_{\text{YES}}, \Pi_{\text{NO}})$ that can be solved by a *uniform* family of quantum circuits $\{U_n\}$ in the following sense:

Let $\mathbf{x}$ be a binary string of length n . Then the application of U_n to $|\mathbf{x}, 0\rangle$ produces the state

$$U_n|\mathbf{x}, 0\rangle = \alpha_{\mathbf{x},0}|0\rangle \otimes |\psi_{\mathbf{x},0}\rangle + \alpha_{\mathbf{x},1}|1\rangle \otimes |\psi_{\mathbf{x},1}\rangle$$

such that

• $|\alpha_{\mathbf{x},1}|^2 \geq 2/3$ if $x \in \Pi_{\text{YES}}$

• $|\alpha_{\mathbf{x},1}|^2 \leq 1/3$ if $x \in \Pi_{\text{NO}}$

note that nothing is required with respect to inputs that violate the promise

Complete Problems

- no complete problems are known for complexity classes such as BPP, BQP, and MA
- complete problems are only known for their promise versions

String Rewriting Problem

- let $\mathcal{A}$ be a finite alphabet
- let $\sim$ be a symmetric relation on substrings of length at most k (where k is a constant)
- let A be the adjacency matrix of the undirected graph $G = (V, E)$, where
 - the set of vertices V is $\mathcal{A}^r$ (strings of length r over $\mathcal{A}$)
 - the set of edges E corresponds to possible conversions between strings, that is,

$$(avb, awb) \in E \iff v \sim w$$

String Rewriting Problem

• let s , t , and t' be three strings in $\mathcal{A}^r$

• let

$$\Delta_{s,t,t'}(n) := A_{s,t}^n - A_{s,t'}^n$$

be the difference between the number of possibilities to obtain t from s and those to obtain t' from s after exactly n replacements

• given the promise

• $|\Delta_{s,t,t'}(n)| \leq c^n$ for all $n \in \mathbb{N}$ (*growth condition*)

• $|\Delta_{s,t,t'}(m)| \geq \epsilon c^m$

determine the sign of $\Delta_{s,t,t'}(m)$ where

$c > 0$ is a constant, $m \in \mathbb{N}$ with $m = \text{poly}(r)$, and $\epsilon \in [0, 1]$
with $\epsilon = \text{poly}(1/r)$

Motivation from Bioinformatics

- strings s , t , and t' correspond to DNA sequences
- mutations change substrings (possible mutations are specified by the relation $\sim$)
- String rewriting problem:
Are there more possibilities to mutate from s to t or from s to t' if exactly m mutations occur?

PromiseBQP-complete

Theorem: String Rewriting is PromiseBQP-complete

The string rewriting problem is the most difficult problem that can be efficiently solved on a quantum computer.

Quantum computers are better at estimating differences of combinatorial quantities than classical computers (provided that BPP is not equal to BQP).

String Rewriting is in PromiseBQP

- rewrite $\Delta_{s,t,t'}(m)$ as a difference of expected values

$$\Delta_{s,t,t'}(m) = \sqrt{2}(\langle \psi^+ | A^m | \psi^+ \rangle - \langle \psi^- | A^m | \psi^- \rangle)$$

with $|\psi^\pm\rangle = (|s\rangle \pm |\varphi\rangle)/\sqrt{2}$ and $|\varphi\rangle = (|t\rangle + |t'\rangle)/\sqrt{2}$

- determine the expected values as follows:

- let $U := \exp(-iA/b)$ where $b \leq \|A\|$
- apply quantum phase estimation to U and $|\psi^\pm\rangle$

this allows one to sample according to the spectral measure induced by A and $|\psi^\pm\rangle$:

eigenvalue λ_j occurs with probability $\langle \psi^\pm | P_j | \psi^\pm \rangle$

- raise the outcome to the m th power
- output average value after sufficiently many runs

Technical Difficulties

- imperfect simulated time evolution
- imperfect reproduction of the spectral measure
- amplification of errors due to $x \mapsto x^m$
 $\implies$ use a Lipschitz-continuous function instead
- imperfect estimation due to statistical fluctuations
- most serious problem: the accuracy of the estimation is of order b^m
 $\implies$ the growth condition allows one to zoom onto the right scale, making it possible to attain accuracy of order c^m

String Rewriting is PromiseBQP-hard

Equivalent Definition of PromiseBQP

- let U be a quantum circuit and let $\mathbf{x}$ be the input
- let p_0 and p_1 denote the probability of finding the output qubit in state $|0\rangle$ and $|1\rangle$, respectively
- elementary calculations show that

$$\langle \mathbf{x}, 0 \dots 0 | U^\dagger \sigma_z U | \mathbf{x}, 0 \dots 0 \rangle = p_0 - p_1$$

- the sign of this quantity indicates if $x \in \Pi_{\text{YES}}$ or $x \in \Pi_{\text{NO}}$

Quantum-Universal Interaction

we construct a quantum-universal translationally-invariant finite-range interaction on a line

$$H = F + F^\dagger = \sum_j V_j + \sum_j V_j^\dagger$$

acting on $\mathcal{H} := (\mathbb{C}^d)^{\otimes l}$ such that

$$\langle \omega | H^m | \alpha \rangle = (p_0 - p_1) \cdot w(m)$$

where $|\alpha\rangle$ and $|\omega\rangle$ are basis states of $\mathcal{H}$ that encode the input x and the circuit U and w is a certain function with $w : \mathbb{N} \rightarrow \mathbb{R}_+$

$\implies$ the sign of $\langle \omega | H^m | \alpha \rangle$ indicates if $x \in \Pi_{\text{YES}}$ or $x \in \Pi_{\text{NO}}$

Action of Forward-Time Operator

#	#	#	#	#	⊞	I	T	I	H	I	H	S	I	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	■	I	T	I	H	I	H	S	I	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	I	■	T	I	H	I	H	S	I	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	I	T	■	I	H	I	H	S	I	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

■ moves to the end of the program code

Action of Forward-Time Operator

#	#	#	#	#	I	T	I	H	I	H	S	I	■	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	I	T	I	H	I	H	S	I	◇	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	I	T	I	H	I	H	S	II	#	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of Forward-Time Operator

#	#	#	#	#	I	T	I	H	I	H	S	I	#	#	#
	•	•	•	•		b_1	b_2	b_3	b_4	•	•	•	•	•	•

Action of H on the Relevant Subspace

- the restriction of the Hamiltonian H to the span of $F^i|\alpha\rangle$ is unitarily equivalent to the adjacency matrix of the line graph

$$\mathcal{P} := \sum_{j=0}^{\ell-1} |j\rangle\langle j+1| + |j+1\rangle\langle j|$$

where ℓ is the number of transitions until the program code has moved all the way to the left (so that no transition is possible anymore)

$$\langle\omega|H^m|\alpha\rangle = (p_0 - p_1)(\mathcal{P}^m)_{0,\ell-1}$$

Hamiltonian H

$$H = F + F^\dagger = \sum_{j=1}^r V_j + V_j^\dagger$$

the operator V acts on three adjacent cells

$$V = Z(U_T P_T + U_S P_S + U_H P_H + P_I)$$

- Z defines the annihilation and creation of configurations
- P_G is a projector onto $|G\rangle$ (indicating that gate G has to be applied)
- U_G implements the gate G

Conversion of H into string rewriting rules

- the entries of the Hamiltonian H are $0, 1, \pm 1/\sqrt{2}$
- the entries should be only 0 and 1, so that the Hamiltonian defines an adjacency matrix
- it is possible to construct a Hamiltonian $\tilde{H}$ such that it contains only entries $0, \pm 1/\sqrt{2}$
- $\tilde{H}$ has the same spectral properties as H

Conversion into Adjacency Matrix

- the operator $\tilde{V}$ acts on three cells $(\mathbb{C}^d)^{\otimes 3}$

$$\tilde{V} = \sum_{a,b} s_{ab} |a\rangle\langle b|$$

- define operators X_{ab} acting on $(\mathbb{C}^2)^{\otimes 3}$ ('minus-sign simulator' cells) by setting

$$X_{ab} = I \otimes \sigma_x \otimes I \quad \text{if} \quad s_{ab} = -1/\sqrt{2}$$

$$X_{ab} = I \otimes I \otimes I \quad \text{if} \quad s_{ab} = 1/\sqrt{2}$$

$$X_{ab} = 0 \quad \text{if} \quad s_{ab} = 0$$

Conversion into Adjacency Matrix

- prepare the simulator cells in the state

$$|\psi\rangle = |0\rangle \otimes |-\rangle \otimes |0\rangle$$

- then we can simulate the minus sign since

$$X_{ab}|\psi\rangle = s_{ab}|\psi\rangle$$

- define the operator $\hat{V}$ by setting

$$\hat{V} = \sqrt{2} \sum_{a,b} X_{ab} \otimes |a\rangle\langle b|$$

- $\hat{V}$ defines the substitution rules for substrings

- let A be the corresponding Hamiltonian/adjacency matrix

Conversion into String Rewriting Problem

- one can show that there are three strings $s := \alpha_0$, $t := \omega_0$, and $t' := \omega_1$ such that

$$\begin{aligned}(A^m)_{st} - (A^m)_{st'} &= \langle \omega_0 | A^m | \alpha_0 \rangle - \langle \omega_1 | A^m | \alpha_0 \rangle \\ &= (p_0 - p_1) \cdot \sqrt{2}^m \cdot (\mathcal{P}^m)_{0,\ell-1}\end{aligned}$$

- eigenvalues of the line graph are in $[-2, 2]$

$\Rightarrow$ growth condition is satisfied with $c := 2\sqrt{2}$

- one can choose $m = \text{poly}(r)$ such that

$$\sqrt{2}^m \cdot (\mathcal{P}^m)_{0,\ell-1} = 1/\text{poly}(r) \cdot c^m$$

$\Rightarrow$ it suffices to estimate $\Delta_{s,t,t'}(m)$ with precision ϵc^m (where $\epsilon = 1/\text{poly}(r)$) to determine the sign

Conclusions

- a purely classical (quantum free) characterization of BQP
- quantum computers are better at counting and comparing the number of solutions of combinatorial problems (provided that BQP is strictly larger BQP)